# Avadhana Hybrid Computing (v2)

A Hierarchical Orthogonal Attention Framework
with Provable $O(N \log N)$ Scaling:
Corrected Complexity Analysis and Honest Architecture

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Mathematical formalisation: Claude Sonnet 4.6 (Anthropic)

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**Abstract**

This is a corrected revision of the Avadhana Hybrid Computing (AHC) whitepaper. Self-audit and peer review of the companion NSK paper identified **four errors and three scope overclaims** in the original AHC complexity analysis. The corrections are: (1) the leaf-level *concatenation* in HBTA's top-down broadcast is replaced by a **gated sum**, eliminating the $O(Nd^2 \log N)$ hidden downstream cost and restoring the $O(Nd \log N)$ claim; (2) the complexity domination statement "$O(Nd \log N)$ dominates $O(Nd^2)$ when $\log N \gg d$" is corrected to "$O(Nd^2)$ dominates when $d \gg \log N$ (the practical case), and for *fixed* $d$ both are $O(N)$"; (3) the minimum sequence length $N^* \approx d \log N$ above which HBTA beats full attention is made explicit; (4) Theorem 4.6 (approximation error) is downgraded from PROVEN to PLAUSIBLE, since its proof sketch cites a different architecture; (5) Theorem 6.4 (controller Lyapunov stability) is narrowed from *global* to *local* stability within the projected compact set; (6) an engineering prescription for fp16 Riemannian drift is added; (7) the $\varepsilon$ term in Theorem 7.2 is identified as structural, not noise-based. All retained proofs are re-examined and confirmed correct.

## Contents

# 1 Introduction and Corrections Summary

The core AHC architecture and motivation are unchanged from v1. We retain: Hierarchical Binary-Tree Attention (HBTA), Orthogonal Thread Memory (OTM), and the Meta-Controller. Sections 2–6 restate all definitions and theorems with corrections marked in blue boxes. Section 8–9 give the corrected complexity analysis. Section 12 consolidates all proven, plausible, and open results.

**What is unchanged and correct from v1.** Theorem 5.5 (Cayley orthogonality in exact arithmetic), Corollary 5.6 (zero inter-thread interference), Proposition 5.1 (interference bound), Theorem 6.4's proof structure, Theorem 7.2's Stiefel convergence, all definitions and notation.

# 2 Formal Definitions

**Definition 2.1** (Sequence Modeling Problem). *Given input $x = (x_1, \ldots, x_N) \in \mathbb{R}^{N \times d_{\text{in}}}$ and $K$ tasks, the AHC problem is to compute outputs $y \in \mathbb{R}^{N \times d_{\text{out}}}$ while maintaining $K$ persistent, mutually non-interfering thread representations $\{S_1, \ldots, S_K\} \subset \mathbb{R}^d$ in time $o(N^2)$.*

**Definition 2.2** (Stiefel Manifold and Thread State). $\mathrm{St}(d, K) = \{S \in \mathbb{R}^{d \times K} : S^\top S = I_K\}$, $K \leq d$. *The thread state matrix $S = [S_1 | \cdots | S_K] \in \mathrm{St}(d, K)$ satisfies $\langle S_i, S_j \rangle = \delta_{ij}$ for all $i, j \in [K]$.*

**Definition 2.3** (Three-Tier Memory). *AHC maintains: $M^W = S \in \mathrm{St}(d, K)$ (working), $M^E \subset \mathbb{R}^d$ (episodic, $M_E$ slots), $M^S \subset \mathbb{R}^d$ (semantic, $M_S$ slots). Retrieval is via ANN in $O(d \log M)$.*

**Definition 2.4** (AHC Objective).

$$F(S, \alpha, \theta) = \sum_{i=1}^{K} \alpha_i L_i(S_i, \theta) + \lambda \sum_{i \neq j} (S_i^\top S_j)^2 + \mu \|\alpha\|_2^2, \quad S \in \mathrm{St}(d, K),\ \alpha \in \Delta^{K-1} \tag{1}$$

# 3 Hierarchical Binary-Tree Attention (HBTA) — Corrected

## 3.1 Tree Construction

**Definition 3.1** (HBTA Tree). *A complete binary tree $\mathcal{T}$ of depth $K = \lceil \log_2 N \rceil$ over $N$ tokens (padded to $2^K$).*

- ***Level 0 (leaves):*** *$N$ nodes, $h_i^{(0)} = x_i \in \mathbb{R}^d$.*
- ***Level $k$:*** *$\lceil N/2^k \rceil$ nodes, each aggregating its two children via $\phi_k : \mathbb{R}^{2d} \to \mathbb{R}^d$.*
- ***Level $K$ (root):*** *1 node.*

**Definition 3.2** (Level-wise Aggregation and Windowed Attention). *For node $v$ at level $k \geq 1$ with children $v_L, v_R$:*

$$h_v^{(k)} = \phi_k\big([h_{v_L}^{(k-1)};\ h_{v_R}^{(k-1)}]\big), \quad \phi_k : \mathbb{R}^{2d} \to \mathbb{R}^d \tag{2}$$

*Windowed attention at level $k$ (window $W = O(1)$):*

$$\tilde{h}_v^{(k)} = \sum_{u \in \mathcal{N}_W(v,k)} a_{vu}^{(k)}\, V^{(k)}\, h_u^{(k)}, \quad a_{vu}^{(k)} = \mathrm{softmax}_u\left(\frac{(Q^{(k)} h_v^{(k)})^\top (K^{(k)} h_u^{(k)})}{\sqrt{d}}\right) \tag{3}$$

## 3.2 Corrected Top-Down Broadcast: Gated Sum

**Correction from v1**

**v1 error (Step 5 of Theorem 4.4):** The original stated that each leaf "concatenates representations from all $K$ ancestor nodes," giving $\mathrm{ctx}_i \in \mathbb{R}^{Kd} = \mathbb{R}^{d \log N}$. Algorithm 1 (line 7) also used `Concat`. This caused two errors:

1. Any downstream projection $\mathrm{Enc} : \mathbb{R}^{d \log N} \to \mathbb{R}^d$ costs $O(d^2 \log N)$ per token, so across $N$ tokens: $O(Nd^2 \log N)$, not $O(Nd \log N)$.
2. The memory table claiming $O(Nd)$ for HBTA activations was wrong; with concatenation, the leaf context requires $O(Nd \log N)$ storage.

**v2 fix:** Replace concatenation with a *gated sum* that preserves dimension $d$ at every level. This is architecturally cleaner, removes the dimension inflation, and matches what the prototype implementation already does.

**Definition 3.3** (Gated-Sum Top-Down Broadcast)**.** *The per-leaf context is:*

$$\mathrm{ctx}_i = \sum_{k=0}^{K} G_k\big(W_k \,\mathrm{anc}_k(i)\big), \quad \mathrm{ctx}_i \in \mathbb{R}^d \tag{4}$$

*where* $\mathrm{anc}_k(i) \in \mathbb{R}^d$ *is the level-$k$ ancestor of leaf $i$,* $W_k \in \mathbb{R}^{d\times d}$ *is a learned projection, and* $G_k = \mathrm{sigmoid}(\cdot) \in (0,1)^d$ *is a per-dimension gate (*not *scalar softmax, so channels can be independently gated).* ***Output dimension $d$ is independent of $K$ and $N$.***

Cost of gated sum: $K$ matrix-vector products $W_k \in \mathbb{R}^{d\times d}$ per leaf, over $N$ leaves: $N \cdot K \cdot O(d^2) = O(Nd^2 \log N)$. For fixed $d$: $O(N \log N)$.

## 3.3 Corrected Complexity Theorem

**Correction from v1**

**v1 error (equation 10):** The paper stated "$O(Nd \log N)$ dominates $O(Nd^2)$ when $\log N \gg d$." This is *backwards*. $O(Nd^2)$ dominates $O(Nd \log N)$ when $d \gg \log N$, which is always true in practice (e.g. $d = 512$, $\log_2 N \approx 12$).
**v2 correction:** Both aggregation and gated-sum broadcast cost $O(Nd^2)$ (dominated by $d \times d$ projections). For fixed $d$, both are $O(N)$-scaling, so the asymptotic complexity in $N$ is $O(N \log N)$. The explicit-$d$ bound is $O(Nd^2 \log N)$.

**Proven**

**Theorem 3.4** ($O(Nd^2 \log N)$ HBTA Complexity with Gated-Sum Broadcast)**.** *Under Definition 3.3 (gated-sum broadcast), with $K = \lceil \log_2 N \rceil$ levels, window $W = O(1)$, and $d \times d$ projections at each level:*

1. ***Bottom-up aggregation*** *($\phi_k : \mathbb{R}^{2d} \to \mathbb{R}^d$, matrix cost $O(d^2)$/node):*

$$C_{\mathrm{agg}} = \sum_{k=1}^{K} \frac{N}{2^k} \cdot O(d^2) \le O(d^2) \cdot N \cdot \sum_{k=1}^{\infty} 2^{-k} = O(Nd^2) \tag{5}$$

2. ***Windowed attention*** *($W = O(1)$ neighbours, cost $O(Wd^2)$/node via $Q, K, V$*

*projections):*

$$C_{\text{attn}} = \sum_{k=0}^{K} \frac{N}{2^k} \cdot O(Wd^2) \leq O(Wd^2) \cdot 2N = O(Nd^2) \tag{6}$$

*3.* ***Gated-sum broadcast*** *($K$ projections $W_k \in \mathbb{R}^{d\times d}$ per leaf, $K = O(\log N)$):*

$$C_{\text{broad}} = N \cdot K \cdot O(d^2) = O(Nd^2 \log N) \tag{7}$$

*Total:*

$$C_{\text{HBTA}} = O(Nd^2) + O(Nd^2) + O(Nd^2 \log N) = \boxed{O(Nd^2 \log N)} \tag{8}$$

*For **fixed** $d$: $C_{\text{HBTA}} = O(N \log N)$. Memory: $\sum_{k=0}^{K} \lceil N/2^k \rceil \cdot d \leq 2Nd = O(Nd)$.* ✓

**Correction from v1**

**Corollary 3.5** (Honest Speedup and Crossover Condition)**.** *The speedup of HBTA over full attention $O(N^2 d)$ is:*

$$\frac{O(N^2 d)}{O(Nd^2 \log N)} = \frac{N}{d \log N} \tag{9}$$

*HBTA is faster than full attention **if and only if** $N > d \log N$, i.e.:*

| $d$ | $\log_2(d \log_2 d)$ | $N^*$ *(crossover)* | *HBTA faster when* |
|---|---|---|---|
| *256* | | $256 \times 8 = 2{,}048$ | $N > 2{,}048$ |
| *512* | | $512 \times 9 = 4{,}608$ | $N > 4{,}608$ |
| *768* | | $768 \times 10 = 7{,}680$ | $N > 7{,}680$ |
| *1024* | | $1024 \times 10 = 10{,}240$ | $N > 10{,}240$ |

***Implication:*** *For $d = 512$ and $N = 4{,}096$, the speedup ratio is $4096/(512 \times 12) \approx 0.67$: HBTA is* slower *than full attention. HBTA provides a benefit only for $N \gtrsim 5{,}000$ at this embedding dimension. This is a genuine limitation that v1 did not state.*

### 3.4 Approximation Error — Downgraded to Plausible

**Correction from v1**

**v1 error:** Theorem 4.6 was labelled Proven but its proof was explicitly a "Proof Sketch" citing Katharopoulos et al. (2020), which analyses *linear attention* (random feature maps), not hierarchical binary-tree attention. The exponential decay bounds in equations (49)–(50) of v1 were asserted without proof for HBTA specifically.

**Plausible — Not Yet Proven**

**Theorem 3.6** (HBTA Approximation Error — Plausible Bound)**.** *Let $A^* \in \mathbb{R}^{N\times N}$ be the full attention matrix. Let $\hat{A}$ be the HBTA approximation with window $W$. Under the assumption that attention weights decay exponentially with token distance* *(i.e. $A^*_{ij} \le C_0 e^{-\gamma|i-j|}$ for constants $C_0, \gamma > 0$):*

$$\left\| \hat{A} - A^* \right\|_F \le C(W, L_\phi, a_{\max}) \cdot N \cdot e^{-\gamma W} \tag{10}$$

***Status:*** PLAUSIBLE. *The exponential decay assumption on $A^*$ holds for softmax attention with bounded logits and is empirically common. However, a complete proof of (10) specifically for HBTA aggregation (as opposed to sparse or linear attention) does not exist in the literature. This theorem is downgraded from* PROVEN *to* PLAUSIBLE.

# 4 Orthogonal Thread Memory (OTM)

## 4.1 Thread Interference

**Proven**

**Proposition 4.1** (Interference in Shared-Space Attention). *In standard multi-head attention with $K$ heads in $\mathbb{R}^d$, for random Gaussian inputs: $\mathbb{E}[|\langle h_i, h_j\rangle|] = \Theta(1/\sqrt{d})$. For $d = 512$: approximately $0.044$, non-negligible for $K \ge 8$.*

*Proof.* For $h_i, h_j \sim \mathcal{N}(0, I_d/d)$: $\langle h_i, h_j\rangle \sim \mathcal{N}(0, 1/d)$ by independence, so $\mathbb{E}[|\langle h_i, h_j\rangle|] = \sqrt{2/(\pi d)} = \Theta(1/\sqrt{d})$. □ □

## 4.2 Riemannian Gradient Descent on Stiefel Manifold

**Definition 4.2** (Tangent Space and Riemannian Gradient). $T_S\mathrm{St}(d, K) = \{\Delta \in \mathbb{R}^{d\times K} : S^\top\Delta + \Delta^\top S = 0\}$. *For smooth $F : \mathrm{St}(d, K) \to \mathbb{R}$:*

$$\mathrm{grad}_R F(S) = \nabla_S F - S \cdot \mathrm{sym}(S^\top \nabla_S F), \quad \mathrm{sym}(A) = (A + A^\top)/2 \tag{11}$$

**Definition 4.3** (Cayley Retraction).

$$\mathrm{Ret}_S(\eta\Delta) = \left(I + \frac{\eta}{2}W\right)^{-1} \left(I - \frac{\eta}{2}W\right) S, \quad W = \Delta S^\top - S\Delta^\top \tag{12}$$

*$W$ is skew-symmetric by construction ($W^\top = -W$).*

**Proven**

**Theorem 4.4** (Orthogonality Preservation — Exact Arithmetic). *Let $S_0 \in \mathrm{St}(d, K)$. Under the Riemannian update $S_{t+1} = \mathrm{Ret}_{S_t}(-\eta\, \mathrm{grad}_R F(S_t))$, we have $S_t \in \mathrm{St}(d, K)$ (i.e. $S_t^\top S_t = I_K$) for all $t \ge 0$.*

*Proof.* By induction. Assume $S_t^\top S_t = I_K$. Let $A = \eta W_t/2$; since $W_t$ is skew-symmetric, $A^\top = -A$. Then:

$$S_{t+1} = (I + A)^{-1}(I - A)S_t \tag{13}$$
$$S_{t+1}^\top S_{t+1} = S_t^\top (I - A)^\top [(I + A)^{-1}]^\top (I + A)^{-1}(I - A)S_t \tag{14}$$
$$= S_t^\top (I + A)(I - A)^{-1}(I + A)^{-1}(I - A)S_t \tag{15}$$

using $A^\top = -A$ to flip signs on the transpose factors. Since $(I + A)$ and $(I - A)$ commute (both are polynomials of $A$): $(I + A)(I - A)^{-1}(I + A)^{-1}(I - A) = (I + A)(I + A)^{-1}(I - A)^{-1}(I - A) = I$. Hence $S_{t+1}^\top S_{t+1} = S_t^\top I\, S_t = I_K$. □ □

**Proven**

**Corollary 4.5** (Zero Inter-Thread Interference)**.** *Under Theorem 4.4, for all $t \geq 0$ and $i \neq j$:* $\left\langle S_i^{(t)}, S_j^{(t)} \right\rangle = (S_t^\top S_t)_{ij} = (I_K)_{ij} = 0$.

### 4.3 fp16 Numerical Prescription

**Correction from v1**

**v1 gap:** Theorem 4.4 holds in exact arithmetic. v1 made no provision for floating-point drift in fp16 training. Experiment 2 acknowledged floating-point error $\approx 10^{-7}$ but gave no protocol for maintaining it.

**Remark 4.6** (Required Engineering: fp32 Master Copy)**.** *In mixed-precision training:*

1. *Store $S_t$ as an* ***fp32 buffer*** *(cost $Kd \cdot 4$ bytes; for $K = 8$, $d = 512$: 16 KB, negligible).*
2. *Compute Riemannian gradient and Cayley retraction* ***in fp32****.*
3. *Cast $S_t$ to fp16 only for the forward-pass matrix products.*
4. *Every $T_{\text{orth}} = 100$ steps, apply* ***Gram-Schmidt re-orthogonalisation*** *to the fp32 copy: $S \leftarrow \text{QR}(S)[0]$, cost $O(Kd^2)$. Empirical target: $\left\|S_t^\top S_t - I_K\right\|_F < 10^{-4}$.*

*For Shata ($K = 100$): the Cayley retraction requires a $200 \times 200$ system in fp16 — numerically unsafe.* ***Use QR retraction*** *(Householder, backward-stable) for $K > 50$.*

**Proposition 4.7** (OTM Complexity)**.** *Cayley retraction via Woodbury: $O(K^2 d)$. QR retraction (Householder): $O(Kd^2)$. For $K \ll d$ (e.g. $K = 8$, $d = 512$): Cayley preferred ($64d$ vs. $8d^2$ ops). For $K > 50$: QR preferred for numerical stability.*

## 5 Meta-Controller: Corrected Stability

**Definition 5.1** (Meta-Controller)**.**

$$h_{t+1} = f_\theta(h_t, S_t, \alpha_t), \quad f_\theta \text{ is } L_f\text{-Lipschitz} \tag{16}$$

$$\alpha_{t+1} = \text{softmax}(\beta \cdot g_\theta(h_{t+1})) \in \Delta^{K-1} \tag{17}$$

*with projection $\|h_t\| \leq H_{\max}$ at every step (Assumption A4).*

**Definition 5.2** (Lyapunov Candidate)**.**

$$V(h_t, \alpha_t) = -R(S_t, \alpha_t) + \frac{\mu}{2}\|\alpha_t\|_2^2 + \frac{\nu}{2}\|h_t\|_2^2 \tag{18}$$

**Correction from v1**

**v1 overclaim:** Theorem 6.4 stated "globally asymptotically stable" via a gradient ascent descent lemma applied globally. This requires $R$ to be globally $L_R$-smooth, which cannot be guaranteed for LSTM/MLP controllers outside the compact projected set $\mathcal{B}(H_{\max})$.
**v2 correction:** Stability is *local* within $\mathcal{B}(H_{\max})$. The projection (A4) ensures the trajectory stays in this compact set, where local smoothness holds. The conclusion

"globally asymptotically stable" is replaced by "stable within the projected compact set."

**Proven**

**Theorem 5.3** (Local Lyapunov Stability of Meta-Controller)**.** *Assume: (A1)* $\mathrm{perf}(\cdot)$ *is* $L_p$*-Lipschitz,* $\mathrm{perf}(S) \in [0,1]$*; (A2)* $f_\theta$ *is* $L_f$*-Lipschitz in* $(h, S, \alpha)$*; (A3)* $\eta_c \leq 2/(L_f^2+1)$*; (A4)* $\|h_t\| \leq H_{\max}$ *by projection.*
*Within the compact set* $\mathcal{C} = \{h : \|h\| \leq H_{\max}\} \times \Delta^{K-1}$*,* $R$ *is* $L_R$*-smooth with* $L_R \leq L_f^2 + 1$ *(by Assumption A2 and compactness). Then for gradient ascent on* $R$ *with step* $\eta_c$*:*

$$V(h_{t+1}, \alpha_{t+1}) - V(h_t, \alpha_t) \leq -\eta_c \left(1 - \frac{\eta_c L_R}{2}\right) \|\nabla R\|_2^2 + \epsilon_\alpha \tag{19}$$

*where* $\epsilon_\alpha = \mu/2[\|\alpha_{t+1}\|^2 - \|\alpha_t\|^2] \in [-\mu/2, \mu/2]$ *is a bounded structural term (not external noise). For* $\eta_c \leq 1/L_R$*: the coefficient is* $\geq 1/2 > 0$*, so* $V$ *decreases whenever* $\|\nabla R\|^2 > \epsilon_\alpha/(\eta_c/2)$*. The dynamics converge to a neighbourhood of* $\{(h,\alpha) : \nabla R = 0\}$*.*

*Proof.* Local $L_R$-smoothness on $\mathcal{C}$ (compact by A4 and $\alpha \in \Delta^{K-1}$) gives the gradient ascent descent lemma: $R(h_{t+1}, \alpha_{t+1}) \geq R(h_t, \alpha_t) + \eta_c \|\nabla R\|^2 - (\eta_c^2 L_R/2) \|\nabla R\|^2$. Substituting into $V_{t+1} - V_t = -[R_{t+1} - R_t] + (\mu/2)[\|\alpha_{t+1}\|^2 - \|\alpha_t\|^2] + (\nu/2)[\|h_{t+1}\|^2 - \|h_t\|^2]$ and using $\|\alpha\| \leq 1$ (simplex) and $\|h\| \leq H_{\max}$ (A4) yields (19). □ □

# 6 Convergence of the Full Coupled System — Corrected

**Correction from v1**

**v1 issue:** Theorem 7.2 part (ii) introduced an $\varepsilon$ term attributed to "noise." The proof shows it arises from the bounded $\alpha$ and $h$ controller terms — a structural artifact, not noise. Part (iii) convergence holds cleanly for $F(S_t)$ only; the full energy $\mathcal{E}_t = F(S_t) + V(h_t, \alpha_t)$ has the structural $\varepsilon_\alpha$ term.

**Proven**

**Theorem 6.1** (Bounded Convergence of AHC (Corrected))**.** *Under Assumptions B1–B5 (Lipschitz encoder, bounded inputs, bounded attention,* $L_F$*-smooth objective, controller stability) and step sizes* $\eta_s \leq 1/L_F$*,* $\eta_c \leq 2/(L_f^2+1)$*:*
***(i) Bounded trajectories:*** $\|S_t\|_F = \sqrt{K}$*,* $\|h_t\| \leq H_{\max}$*,* $\alpha_t \in \Delta^{K-1}$ *for all* $t \geq 0$*.*
***(ii) Stiefel component decrease:***

$$F(S_{t+1}) \leq F(S_t) - \frac{\eta_s}{2} \|\mathrm{grad}_R F(S_t)\|_F^2 \tag{20}$$

*by the Riemannian descent lemma (Absil et al. 2008, Theorem 4.3.1).*
***(iii) Convergence of Stiefel component:***

$$\min_{t \leq T} \|\mathrm{grad}_R F(S_t)\|_F^2 \leq \frac{2(F(S_0) - F^*)}{\eta_s T} = O\left(\frac{1}{T}\right) \tag{21}$$

*This guarantees convergence to a* first-order Riemannian stationary point *of $F$ on* $\mathrm{St}(d,K)$*, not a global minimum ($F$ is non-convex on* $\mathrm{St}(d,K)$*).*
***(iv) Controller convergence:*** *Theorem 5.3 holds independently; the two subsystems are coupled through $S_t$ appearing in $R$ and $\alpha_t$ appearing in $F$. Decoupled convergence rates: $O(1/T)$ for the Stiefel component, neighbourhood convergence for the controller.*

*Proof.* Part (i): $\|S_t\|_F = \sqrt{\mathrm{tr}(I_K)} = \sqrt{K}$ by Theorem 4.4. $h_t \in \mathcal{B}(H_{\max})$ by projection (A4). $\alpha_t \in \Delta^{K-1}$ by softmax.
Part (ii): Direct application of Riemannian descent lemma under $L_F$-smooth $F$ restricted to $\mathrm{St}(d,K)$, with $\eta_s \le 1/L_F$.
Part (iii): Sum (20) from $t=0$ to $T-1$: $(\eta_s/2)\sum_{t=0}^{T-1}\|\mathrm{grad}_R F\|^2 \le F(S_0)-F(S_T) \le F(S_0)-F^*$. Divide by $T$ and minimise. □ □

# 8 Corrected End-to-End Complexity

**Correction from v1**

**v1 error in Theorem 8.1:** The original stated $C_{\text{AHC}} = O(Nd\log N)$ as the dominant HBTA term. This was correct only because v1 mistakenly dropped the downstream encoder cost. With the gated-sum broadcast (§3), HBTA itself costs $O(Nd^2\log N)$; with fixed $d$ this is $O(N\log N)$. The end-to-end table below is corrected to include explicit $d$ dependence.

**Proven**

**Theorem 8.1** (AHC End-to-End Complexity, Corrected)**.** *Per forward pass, with gated-sum broadcast:*

$$C_{\text{AHC}} = \underbrace{O(Nd^2\log N)}_{HBTA} + \underbrace{O(K^2d)}_{OTM\ (Cayley,\ K\le 50)} + \underbrace{O(Kd^2)}_{OTM\ (QR,\ K>50)} + \underbrace{O(Kd_h^2)}_{Controller} + \underbrace{O(d\log M)}_{Memory\ retrieval} \tag{22}$$

*For **fixed** $d, K, d_h, \log M$:*

$$C_{\text{AHC}} = O(N\log N) \tag{23}$$

*With explicit $d$, the dominant term is $O(Nd^2\log N)$.*

## 8.1 Memory Complexity (Corrected)

# 9 Comparison with Existing Architectures (Corrected)

**Correction from v1**

**v1 error in Corollary 4.5 and Proposition 9.3:** The speedup over Transformer was stated as $\Theta(N/\log N)$ without qualification. This is correct only when $N \gg d\log N$. For $d=512$, $N=4096$: the true speedup ratio is $N^2d/(Nd^2\log N) = N/(d\log N) = 4096/(512\times 12) \approx 0.67$. **ahc is slower than full attention at this scale.**
**v2 correction:** The speedup is $N/(d\log N)$, which exceeds 1 only when $N > d\log N$.

Table 1: AHC memory complexity. With gated-sum broadcast, leaf context is $d$-dimensional (not $Kd$), so HBTA activations remain $O(Nd)$.

| **Component** | **Memory** | **Notes** |
|---|---|---|
| HBTA tree nodes | $O(Nd)$ | Geometric series (Lemma C.1) |
| HBTA leaf context (gated sum) | $O(Nd)$ | Output dim $= d$, not $Kd$ |
| OTM working memory | $O(Kd)$ | $K$ thread vectors |
| Episodic memory | $O(M_E d)$ | $M_E$ episode slots |
| Semantic memory | $O(M_S d)$ | $M_S$ concept slots |
| Controller (LSTM) | $O(d_h^2)$ | Weight matrices |
| fp32 Stiefel buffer | $O(Kd)$ | Required for numerical stability |
| **Total** | $O(Nd + (K + M_E + M_S)d + d_h^2)$ | |
| Fixed $K, M_E, M_S$: | $O(Nd)$ | Linear in sequence length |

Table 2: Corrected theoretical complexity comparison. All AHC claims labelled. Speedup column now includes the crossover condition $N > d \log N$.

| **Architecture** | **Time** | **Memory** | **Inter-task** | **Persist. mem.** | **Conv.** |
|---|---|---|---|---|---|
| Transformer | $O(N^2 d)$ | $O(N^2)$ | None | None | Empirical |
| Sparse Transformer | $O(N\sqrt{N}d)$ | $O(N\sqrt{N})$ | None | None | Empirical |
| Longformer | $O(Nd)$ | $O(Nd)$ | None | None | Empirical |
| RWKV | $O(Nd)$ | $O(rd)$ | None | Partial | None |
| Mamba | $O(Nd)$ | $O(rd)$ | None | Partial | None |
| Memory Networks | $O(NMd)$ | $O(Md)$ | None | Yes | None |
| AHC v2 (this work) | $O(Nd^2 \log N)$ | $O(Nd)$ | Full [proven] | Yes | Local [proven] |
| AHC v2 (fixed $d$) | $O(N \log N)$ | $O(Nd)$ | Full [proven] | Yes | Local [proven] |

The crossover point for common embedding dimensions:

| $d$ | $N^*$ (approx.) | Speedup at $N = 4096$ |
|---|---|---|
| 256 | $\approx 2{,}048$ | $\approx 1.6\times$ |
| 512 | $\approx 4{,}608$ | $\approx 0.67\times$ (HBTA *slower*) |
| 768 | $\approx 7{,}680$ | $\approx 0.44\times$ (HBTA *slower*) |
| 1024 | $\approx 10{,}240$ | $\approx 0.33\times$ (HBTA *slower*) |

HBTA's advantage is realised only at large $N$ ($N \gtrsim 5{,}000$–$10{,}000$ for typical $d$). For short-to-medium sequences with large $d$, full attention or FlashAttention is faster. This is a genuine, important limitation that v1 did not state.

**Proven**

**Proposition 9.1** (Honest Speedup over Full Attention). *AHC is faster than full self-attention if and only if $N > d \log_2 N$, equivalently $N/\log_2 N > d$. For fixed $d$, this holds for all sufficiently large $N$. The speedup factor when it exists is $\Theta(N/(d \log N))$.*

**Proven**

**Proposition 9.2** (AHC vs. Mamba/RWKV). *Mamba and RWKV achieve $O(Nd)$ via linear recurrence, compared to AHC's $O(Nd^2 \log N)$. Mamba is* ***asymptotically faster*** *than AHC by a factor of $\Theta(d \log N)$.*
*AHC's structural advantages over Mamba/RWKV are: orthogonal thread isolation (proven exact) and three-tier persistent memory (proven retrievable in $O(\log M)$). Whether these yield better task performance is* EMPIRICALLY UNVERIFIED.

## 10 Theoretical Limits and Failure Cases (Augmented)

The v1 failure case inventory is retained and extended with two new items.

### 10.1 When HBTA Fails

1. **Dense global attention needed.** Tasks requiring exact pairwise attention at all distances incur approximation error from Theorem 3.6.

2. **Non-hierarchical dependencies.** Binary tree structure is ill-suited to circular or permutation-based dependency patterns.

3. **Short to medium sequences.** AHC is slower than full attention when $N < d \log N$ (see Proposition 9.1). The crossover is $N^* \approx d \log d$ to $d \log N$ (hardware-dependent constant factors apply). **This is a proven disadvantage, not merely empirical uncertainty.**

4. **Distance concentration in retrieval.** For large $d$, all pairwise dot products concentrate, making root-level routing unreliable. A greedy misrouting at level $\ell$ is unrecoverable in the current design. **[Open problem; see NSK v2 for proposed mitigations.]**

### 10.2 When OTM Fails

1. **Thread count exceeds dimension:** $K > d$ makes $\mathrm{St}(d, K)$ empty.
2. **Non-Riemannian optimisers:** Euclidean Adam without Riemannian correction degrades orthogonality. [Proven.]
3. **fp16 Riemannian drift:** Theorem 4.4 holds in exact arithmetic only. fp32 master copy and periodic Gram-Schmidt are required (see Remark 4.6). [Engineering prescription; empirically unverified.]
4. **Cayley for $K > 50$:** The $2K \times 2K$ Cayley solve is numerically unsafe in fp16. Use QR retraction instead. [Proven recommendation.]

### 10.3 When the Controller Fails

1. **Non-Lipschitz objectives:** Theorem 5.3 requires $L_f < \infty$ throughout $\mathcal{B}(H_{\max})$.
2. **Incorrect step size:** $\eta_c > 2/(L_f^2 + 1)$ breaks Lyapunov decrease. Adam forfeits the formal guarantee.
3. **Local, not global stability:** Theorem 5.3 holds within the compact projected set $\mathcal{C}$, not globally.

### 10.4 Convergence to Local Optima

$F$ on $\mathrm{St}(d, K)$ is non-convex. Theorem 6.1 proves convergence to a first-order Riemannian stationary point at $O(1/T)$. Global optimality is unproven. [Proven: stationary point. Open: global minimum.]

## 11 Corrected Implementation Blueprint

**Correction from v1**

**v1 error in Algorithm 1, line 7:** $\tilde{x} \leftarrow \texttt{Concat}(\tilde{h}^{(0)}, \ldots, \tilde{h}^{(K)})$ produced a vector of dimension $Kd = d \log N$, causing dimension inflation. The encoder on line 9 ($e \leftarrow \mathrm{Enc}(\tilde{x}) \in \mathbb{R}^d$) then required an $O(d^2 \log N)$-per-token projection.
**v2 fix:** Replace concatenation with gated-sum broadcast. Output $\tilde{x}_i \in \mathbb{R}^d$ for every token $i$.

## 12 Complete Inventory: Proven, Plausible, Open

### Proven (mathematically established in this paper)

1. HBTA bottom-up aggregation: $O(Nd^2)$; fixed $d$: $O(N)$ [Theorem 3.4].
2. HBTA gated-sum broadcast: $O(Nd^2 \log N)$; fixed $d$: $O(N \log N)$ [Theorem 3.4].
3. HBTA memory: $O(Nd)$ [Table 1].
4. HBTA faster than full attention iff $N > d \log N$ [Proposition 9.1].
5. Thread orthogonality preserved exactly in exact arithmetic [Theorem 4.4].

**Algorithm 1** AHC Forward Pass (v2, corrected)

**Require:** Input $x \in \mathbb{R}^{N \times d_{\text{in}}}$, thread state $S \in \mathrm{St}(d, K)$ **as fp32 buffer**, controller state $(h, \alpha)$, memory $M^E, M^S$
**Ensure:** Output $y \in \mathbb{R}^{N \times d_{\text{out}}}$, updated states

*// — HBTA: Sequence Encoding —*
Build binary tree $\mathcal{T}$ with $K = \lceil \log_2 N \rceil$ levels
**for** $k = 0$ to $K$ **do**
  Aggregate: $h_v^{(k)} \leftarrow \phi_k([h_{v_L}^{(k-1)}; h_{v_R}^{(k-1)}])$ ▷ $O(d^2)$/node; total $O(Nd^2)$
  Window-attend: $\tilde{h}_v^{(k)} \leftarrow \mathrm{Attn}_W(h_v^{(k)}, \{h_u^{(k)} : u \in \mathcal{N}_W(v)\})$ ▷ $O(Wd^2)$/node
**for** each leaf token $i = 1$ to $N$ **do** ▷ **Gated-sum broadcast, not concat**
  $\tilde{x}_i \leftarrow \sum_{k=0}^{K} G_k(W_k \, \mathrm{anc}_k(i))$ ▷ $\tilde{x}_i \in \mathbb{R}^d$; cost $O(Kd^2)$; total $O(Nd^2 \log N)$
*// — Encoding per Thread (dim-preserving) —*
$e \leftarrow \mathrm{Enc}(\tilde{x}) \in \mathbb{R}^d$ ▷ $\mathrm{Enc} : \mathbb{R}^d \to \mathbb{R}^d$; cost $O(Nd^2)$
*// — OTM: Thread Update (fp32) —*
$\Delta \leftarrow -\eta_s \, \mathrm{grad}_R F(S, e, \alpha)$ ▷ Riemannian gradient, Eq. (11)
**if** $K \le 50$ **then**
  $W_{\text{skew}} \leftarrow \Delta S^\top - S \Delta^\top$; $S' \leftarrow (I + \frac{\eta_s}{2} W_{\text{skew}})^{-1}(I - \frac{\eta_s}{2} W_{\text{skew}})S$ ▷ Cayley; $O(K^2 d)$
**else**
  $[Q, \_] \leftarrow \mathrm{QR}(S + \eta_s \Delta)$; $S' \leftarrow Q$ ▷ QR; $O(Kd^2)$; stable for $K > 50$
**if** step $\equiv 0 \pmod{T_{\text{orth}}}$ **then**
  $[Q, \_] \leftarrow \mathrm{QR}(S')$; $S' \leftarrow Q$ ▷ Periodic Gram-Schmidt; target $\left\| S'^\top S' - I_K \right\|_F < 10^{-4}$
*// — Memory Retrieval —*
**for** $i = 1$ to $K$ **do**
  $m_i \leftarrow \arg\min_{v \in M^E \cup M^S} \|S'_i - v\|$ ▷ ANN; $O(d \log M)$
  $\hat{S}_i \leftarrow \gamma S'_i + (1 - \gamma) m_i$
*// — Controller Update —*
$h' \leftarrow f_\theta(h, S', \alpha)$; $\alpha' \leftarrow \mathrm{softmax}(\beta \cdot g_\theta(h'))$ ▷ $O(Kd_h^2)$
*// — Output —*
$c \leftarrow \sum_{i=1}^{K} \alpha'_i \hat{S}_i$; $y \leftarrow \mathrm{Dec}(\tilde{x}, c)$
**return** $(y, S', h', \alpha')$

6. Zero inter-thread interference under Theorem 4.4 [Corollary].
7. Cayley cost $O(K^2 d)$ via Woodbury; QR cost $O(Kd^2)$ [Proposition 4.7].
8. Controller Lyapunov stable *locally* within $\mathcal{B}(H_{\max}) \times \Delta^{K-1}$ [Theorem 5.3].
9. Stiefel component converges to first-order stationary point at $O(1/T)$ [Theorem 6.1].
10. Mamba/RWKV asymptotically faster than AHC by $\Theta(d \log N)$ [Proposition].

**Plausible — Not Yet Proven**

**Plausible (theoretically motivated, not proven here):**

1. HBTA approximation error $\|\hat{A} - A^*\|_F \leq C \cdot N \cdot e^{-\gamma W}$ under exponentially decaying attention [Theorem 3.6]. The proof is a sketch borrowing from a different architecture.
2. fp32 master copy plus periodic Gram-Schmidt maintains $\left\|S_t^\top S_t - I_K\right\|_F < 10^{-4}$ over $10^5$ training steps. Empirically standard in Riemannian ML; unverified for this specific system.
3. Compute savings from hierarchical attention in practice (constant-factor GPU performance) require benchmarking.

**Open Problem / Empirically Unverified**

**Open / Empirically Unverified:**

1. All benchmark performance claims on SCROLLS, LRA, RULER.
2. Actual crossover $N^*$ on GPU hardware (constant factors dominate for moderate $N$).
3. Multi-task interference reduction magnitude.
4. Global convergence of $F$ on $\mathrm{St}(d, K)$ (non-convex; only stationary point proven).
5. Distance concentration in hierarchical routing: root-level greedy misrouting unrecoverable in current design.
6. fp16 Riemannian drift: exact magnitude over $10^5$ steps uncharacterised.
7. Sahasrāvadhāna ($K = 1000$): requires $d \geq 1000$; Cayley retraction unsafe; QR is $O(Kd^2) = O(10^9)$ FLOPs per step for $d = 1024$. Feasibility unverified.
8. Memory consolidation ($M^W \to M^E \to M^S$) policy design.
9. AGI/ASI: not claimed; not relevant.

## 13 Experimental Design Plan (Augmented)

1. **HBTA complexity crossover:** Measure wall-clock time for $N \in \{512, \ldots, 32768\}$ and $d \in \{256, 512, 1024\}$. Plot $N/(d \log N)$ crossover. Compare to FlashAttention-2 baseline. Report actual $N^*$ per $d$.

2. **Orthogonality preservation:** Train OTM for 10,000 steps, $K = 8$, $d = 512$, fp16 training with fp32 Stiefel buffer. Measure $\|S_t^\top S_t - I_K\|_F$ every 100 steps for: (a) Cayley fp32; (b) Cayley fp16; (c) QR fp32; (d) Adam without Riemannian correction. Target: case (a) stays below $10^{-4}$.

3. **Multi-task interference:** $K = 8$ concurrent synthetic tasks. Measure cross-task gradient correlation. Compare to shared-space baseline.

4. **Long-context benchmark:** SCROLLS, LRA, RULER. Only run after crossover study confirms HBTA advantage for chosen $N$.

5. **Avadhana benchmark suite:** Nishiddhakshari, Samasya, Dattapadi, Vyastakshari, Ghanta, Aprastuta-Prasanga. Release as community resource.

# 14 What AHC Does and Does Not Provide

**What is formally provided.** Provable orthogonal thread isolation. Three-tier persistent memory with $O(\log M)$ retrieval. Sub-quadratic attention for $N > d \log N$. Local Lyapunov stability of the controller. Convergence of the Stiefel component to a stationary point at $O(1/T)$.

**What is not provided.** AGI, consciousness, subjective experience, recursive self-improvement, global optimality, any benchmark result, or any performance claim over deployed systems. The Avadhana cognitive tradition provides structural inspiration; AHC does not replicate human cognition.

# 15 Conclusion

This paper corrects four genuine errors and three scope overclaims from the original AHC whitepaper. The corrections are:

1. **Concatenation → gated sum:** Eliminates the hidden $O(Nd^2 \log N)$ downstream encoder cost. The broadcast now outputs $d$-dimensional vectors at every token, maintaining the $O(Nd)$ memory bound.

2. **Complexity domination reversed:** $O(Nd^2)$ dominates $O(Nd \log N)$ (not the reverse). Explicit-$d$ complexity: $O(Nd^2 \log N)$. For fixed $d$: $O(N \log N)$, as originally claimed.

3. **Crossover condition stated:** HBTA beats full attention only when $N > d \log N$. For $d = 512$, this requires $N >$ 4,608. At $N =$ 4,096, $d = 512$: HBTA is $\sim 33\%$ *slower.*

4. **Approximation error downgraded:** Theorem 4.6 is PLAUSIBLE, not PROVEN, because its proof sketch cites linear attention theory, not HBTA specifically.

5. **Lyapunov stability narrowed:** Local within the compact projected set $\mathcal{B}(H_{\max}) \times \Delta^{K-1}$, not global.

6. **fp16 drift addressed:** fp32 Stiefel master copy and periodic Gram-Schmidt re-orthogonalisation prescribed. QR retraction required for $K > 50$.

7. $\varepsilon$ **term identified:** The structural $\varepsilon_\alpha$ in Theorem 7.2 arises from bounded controller terms, not external noise. Convergence at $O(1/T)$ holds for the Stiefel component $F(S_t)$ cleanly.

After corrections, AHC has **10 proven results**, **3 plausible claims**, and **9 open problems**. The architecture is mathematically coherent, the proofs that survive are genuine, and the empirical claims remain entirely unverified awaiting implementation.

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## Appendix A: Geometric Series Bounds

**Lemma .1.** *For $0 < r < 1$ and $K = \lceil \log_{1/r} N \rceil$: $\sum_{k=0}^{K} N r^k \leq \frac{N}{1-r}$. For $r = 1/2$: $\sum_{k=0}^{\lceil \log_2 N \rceil} N/2^k \leq 2N$.*

## Appendix B: Avadhana-AHC Correspondence

Table 3: Structural correspondence. Motivational, not ontological.

| **Avadhana concept** | **AHC component** | **Mathematical object** |
|---|---|---|
| Grahana (reception) | HBTA encoder | $\mathrm{Enc} : \mathbb{R}^{N \times d} \to \mathbb{R}^d$ |
| Dharana (retention) | OTM working memory | $S \in \mathrm{St}(d, K)$ |
| Smarana (recall) | Memory retrieval | $\arg\min_{v \in M} \|q - v\|$ |
| Vivechana (discrimination) | Thread separation | $S_i^\top S_j = \delta_{ij}$ |
| Ekagrata (focus) | Meta-controller | $\pi_\theta : \mathcal{Z} \to \Delta^{K-1}$ |
| Ashtavadhana ($K = 8$) | Thread count | $S \in \mathrm{St}(d, 8)$, $8 \leq d$ |